

2024
(Session : 2023-25)

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer any **five** questions in which*

Q. No. 1 is compulsory.

1. Write short notes on any **four** of the following :

$$3\frac{1}{2} \times 4 = 14$$

- (a) Explain Lattice vibration.
- (b) Define dipole-moment and Polarizability.
- (c) Write down the applications of superconducting materials.
- (d) State Gibb's paradox.

- (e) Write down the expression of entropy and Pressure in term of Partition function.
- (f) Define Fermi momentum.
2. Define Di-electric, Ferroelectric and Anti-ferroelectric properties of materials. Discuss Landau's theory of phase transition. $6+8 = 14$
3. Explain Meissner effect and show that superconductors are completely diamagnetic. $10+4 = 14$
4. Discuss the various thermodynamic properties of superconducting transitions. 14
5. Obtain an expression of entropy of an ideal gas. 14
6. State and prove the Liouville's theorem. 14
7. What are the properties of Fermi and Bose gases ? Use the expression of partition function of Grand-canonical ensemble, obtain the expression of partition functions for these two gases. $4+10 = 14$

8. What do you mean by Highly Non-degenerate and degenerate Fermi gases ? Obtain the expression of Fermi-energy. $8+6 = 14$

9. Write short notes on any **two** of the following :

$7 \times 2 = 14$

- (a) Clausius-Mosottic equation
- (b) Destruction of superconductivity by magnetic field
- (c) Bose-Einstein condensation
- (d) N-particle graph of interacting particles

